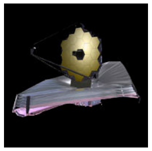

4979 - Uncovering the cold donors of AM CVn binaries

Cycle: 3, Proposal Category: GO

INVESTIGATORS

Name	*Institution*
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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Gaia14aae				
	1	G235M	NIRSpec Bright Object Time Series	(1) Gaia14aae
	2	G395M	NIRSpec Bright Object Time Series	(1) Gaia14aae
ZTFJ1637+49				
	5	G235M	NIRSpec Bright Object Time Series	(3) ZTFJ1637+49
	6	G395M	NIRSpec Bright Object Time Series	(3) ZTFJ1637+49
	7	G395M	NIRSpec Bright Object Time Series	(3) ZTFJ1637+49
SRGeJ0453+62				
	3	G235M	NIRSpec Bright Object Time Series	(2) SRGeJ0453+62
	4	G395M	NIRSpec Bright Object Time Series	(2) SRGeJ0453+62

ABSTRACT

We propose high-cadence NIRSpec observations of three eclipsing, long-period AM CVn binaries using the G235M and G395M gratings. These observations will result in the first-ever detection of AM CVn donors, which are cold (T_eff ~ 1000 K) and only observable in the IR. The main goals of our program are to measure the chemical composition of the donors — which are expected to consist of a mix of helium and metals produced during nuclear burning of their progenitor stars — and to measure their radial velocities, leading to precise constraints on their masses and radii.

These observations will uncover a never-before-observed class of astrophysical object — metal-rich worlds resembling giant planets but consisting mostly of helium — and constrain models for the formation of AM CVn binaries, overcoming a major bottleneck to our understanding of several classes of compact binaries.

OBSERVING DESCRIPTION

We propose to observe each of three targets continuously for ~2 hours total with NIRSpec, with ~1 hour each in the G235M and G395M gratings. We will use the bright object time-series (BOTS) mode with the S1600A1 fixed slit and the SUB2048 subarray. We will use the NRSRAPID readout pattern and 50 groups per integration, for an effective integration time of 46 seconds, yielding ~80 integrations per target per grating. Our targets are compact binaries containing a white dwarf and a low-mass, cold donor. The main goals are to measure radial velocities of the donor over a full orbit (~1 hour) in each grating, and to constrain the atmospheric composition of the donors.

Proposal 4979 - Targets - Uncovering the cold donors of AM CVn binaries

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	Gaia14aae	RA: 16 11 33.9700 (242.8915417d) Dec: +63 08 31.66 (63.14213d) Equinox: J2000	Proper Motion RA: -4.041 mas/yr Proper Motion Dec: -13.888 mas/yr Parallax: 0.0039" Epoch of Position: 2016.0	
Comments: *Category=Star* *Description=[Compact binary stars]* *Extended=NO*				
(2)	SRGeJ0453+62	RA: 04 54 0.1500 (73.5006250d) Dec: +62 24 45.64 (62.41268d) Equinox: J2000	Proper Motion RA: 22.2156 mas/yr Proper Motion Dec: 6.51 mas/yr Parallax: 0.0043" Epoch of Position: 2016.0	
Comments: *Category=Star* *Description=[Compact binary stars]* *Extended=NO*				
(3)	ZTFJ1637+49	RA: 16 37 43.5000 (249.4312500d) Dec: +49 17 40.44 (49.29457d) Equinox: J2000	Proper Motion RA: -46.993 mas/yr Proper Motion Dec: -34.041 mas/yr Parallax: 0.0048" Epoch of Position: 2016.0	
Comments: *Category=Star* *Description=[Compact binary stars]* *Extended=NO*				

Observation

Proposal 4979, Observation 1: G235M

Mon Jun 09 22:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec Bright Object Time Series

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	Gaia14aae	RA: 16 11 33.9700 (242.8915417d) Dec: +63 08 31.66 (63.14213d) Equinox: J2000	Proper Motion RA: -4.041 mas/yr Proper Motion Dec: -13.888 mas/yr Parallax: 0.0039" Epoch of Position: 2016.0	

Comments:
Category=Star
Description=[Compact binary stars]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	WATA	SUB2048	F110W	NRSRAPIDD6	3	1	1	14.452	167835

Template

Subarray
SUB2048

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	G235M/F170LP	NRSRAPID	50	75	1	1	75	3451.686	167835

Special Requirements

Time Series Observation
No Parallel Attachments

Observation

Proposal 4979, Observation 2: G395M

Mon Jun 09 22:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec Bright Object Time Series

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	Gaia14aae	RA: 16 11 33.9700 (242.8915417d) Dec: +63 08 31.66 (63.14213d) Equinox: J2000	Proper Motion RA: -4.041 mas/yr Proper Motion Dec: -13.888 mas/yr Parallax: 0.0039" Epoch of Position: 2016.0	

Comments:
Category=Star
Description=[Compact binary stars]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	WATA	SUB2048	F110W	NRSRAPIDD6	3	1	1	14.452	167835

Template

Subarray
SUB2048

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	G395M/F290LP	NRSRAPID	50	75	1	1	75	3451.686	167835

Special Requirements

Time Series Observation
No Parallel Attachments

Observation

Proposal 4979, Observation 5: G235M

Mon Jun 09 22:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec Bright Object Time Series

Diagnostics

(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	ZTFJ1637+49	RA: 16 37 43.5000 (249.4312500d) Dec: +49 17 40.44 (49.29457d) Equinox: J2000	Proper Motion RA: -46.993 mas/yr Proper Motion Dec: -34.041 mas/yr Parallax: 0.0048" Epoch of Position: 2016.0	

Comments:
Category=Star
Description=[Compact binary stars]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	WATA	SUB2048	F110W	NRSRAPIDD6	3	1	1	14.452	169340

Template

Subarray
SUB2048

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	G235M/F170LP	NRSRAPID	50	88	1	1	88	4049.978	169340

Special Requirements

Time Series Observation
No Parallel Attachments

Observation

Proposal 4979, Observation 6: G395M

Mon Jun 09 22:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec Bright Object Time Series

Diagnostics

(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	ZTFJ1637+49	RA: 16 37 43.5000 (249.4312500d) Dec: +49 17 40.44 (49.29457d) Equinox: J2000	Proper Motion RA: -46.993 mas/yr Proper Motion Dec: -34.041 mas/yr Parallax: 0.0048" Epoch of Position: 2016.0	

Comments:
Category=Star
Description=[Compact binary stars]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	WATA	SUB2048	F110W	NRSRAPIDD6	3	1	1	14.452	169340

Template

Subarray
SUB2048

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	G395M/F290LP	NRSRAPID	50	88	1	1	88	4049.978	169340

Special Requirements

Time Series Observation
No Parallel Attachments

Observation

Proposal 4979, Observation 7: G395M

Mon Jun 09 22:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec Bright Object Time Series

Diagnostics

(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	ZTFJ1637+49	RA: 16 37 43.5000 (249.4312500d) Dec: +49 17 40.44 (49.29457d) Equinox: J2000	Proper Motion RA: -46.993 mas/yr Proper Motion Dec: -34.041 mas/yr Parallax: 0.0048" Epoch of Position: 2016.0	

Comments:
Category=Star
Description=[Compact binary stars]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	WATA	SUB2048	F110W	NRSRAPIDD6	3	1	1	14.452	169340

Template

Subarray
SUB2048

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	G395M/F290LP	NRSRAPID	50	88	1	1	88	4049.978	169340

Special Requirements

Time Series Observation
No Parallel Attachments

Observation

Proposal 4979, Observation 3: G235M

Mon Jun 09 22:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec Bright Object Time Series

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	SRGeJ0453+62	RA: 04 54 0.1500 (73.5006250d) Dec: +62 24 45.64 (62.41268d) Equinox: J2000	Proper Motion RA: 22.2156 mas/yr Proper Motion Dec: 6.51 mas/yr Parallax: 0.0043" Epoch of Position: 2016.0	

Comments:
Category=Star
Description=[Compact binary stars]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	WATA	SUB2048	F110W	NRSRAPIDD6	3	1	1	14.452	169339

Template

Subarray
SUB2048

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	G235M/F170LP	NRSRAPID	50	82	1	1	82	3773.843	169339

Special Requirements

Time Series Observation
No Parallel Attachments

Observation

Proposal 4979, Observation 4: G395M

Mon Jun 09 22:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec Bright Object Time Series

Diagnostics

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	SRGeJ0453+62	RA: 04 54 0.1500 (73.5006250d) Dec: +62 24 45.64 (62.41268d) Equinox: J2000	Proper Motion RA: 22.2156 mas/yr Proper Motion Dec: 6.51 mas/yr Parallax: 0.0043" Epoch of Position: 2016.0	

Comments:
Category=Star
Description=[Compact binary stars]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	WATA	SUB2048	F110W	NRSRAPIDD6	3	1	1	14.452	169339

Template

Subarray
SUB2048

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	G395M/F290LP	NRSRAPID	50	82	1	1	82	3773.843	169339

Special Requirements

Time Series Observation
No Parallel Attachments